Monitoring CO_2 injection for carbon capture and storage using time-lapse 3D VSPs

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Carbon capture and storage (CCS) is the process through which a nearly pure carbon dioxide (CO_2) stream is captured, separated from flue gas or other industrial processes, compressed, transported to an appropriate storage site, and injected deep underground into a geological formation where it can be safely stored for long-term geologic storage (Benson, 2005). Large sedimentary basins, such as the Illinois, Michigan, and Western Canadian sedimentary basins are good targets for CCS, as they are in close proximity to large CO_2 emitters and are composed of the appropriate saline formations and overlying nonpermeable formations. In 2003, the U.S. Department of Energy's National Energy Technology Laboratory (DOE-NETL) created a nationwide network of federal, state, and private sector partnerships to determine the most suitable technologies, regulations, and infrastructure for future CCS in different areas of the North America (Office of Fossil Energy, 2013).

The Illinois Basin–Decatur Project (IBDP), managed by the Midwest Geological Sequestration Consortium (MGSC), is one of the above mentioned partnerships. The MGSC, led by the Illinois State Geological Survey at the University of Illinois, developed IBDP on ADM property in Decatur, Illinois in conjunction with Schlumberger Carbon Services, Archer Daniels Midland Company (ADM), and additional parties. IBDP, which is one of the most advanced U.S. DOE-NETL funded CO_2 storage projects, has a goal to inject one million tons of CO_2 over a three-year period. This project is a fully integrated industrial CCS project where CO_2 from the fermentation process used to produce ethanol at the ADM corn-processing plant is captured instead of being emitted to the atmosphere (Finley et al., 2013). CO_2 injection commenced in November 2011, and as of early August 2013, approximately 562,000 tons of CO_2 had been injected into the basal sands of the Mt. Simon sandstone.

A monitoring, verification, and accounting (MVA) plan is a critical component of a CCS project and has a number of objectives including managing of MVA-related project risks, establishing baseline conditions, demonstrating that project activities are protective of human health and the environment, and providing accurate accounting of stored CO_2 (Locke et al., 2013). Beginning long before injection commenced, extensive site characterization efforts have taken place at the IBDP site that has included 2D and 3D surface seismic surveys as well as well logging and coring programs in the injection (CCS1) and verification (VW1) well (Figure 1). A range of MVA tools are being used to monitor CO_2 from the surface and in the deep subsurface at IBDP, which include time-lapse RST reservoir saturation tool logging in the project wells, continuous pressure and temperature measurements from multiple levels above, within, and below the storage formation, deep fluid sampling with associated geochemical analysis, and seismic monitoring. The primary seismic technologies used in the MVA plan include time-lapse 3D vertical seismic profiles (VSP) and microseismic monitoring. The MVA data is used to calibrate the reservoir simulations on a regular basis, and the project now has robust, history-matched reservoir simulations.

Illinois Basin geology

The Mt. Simon sandstone is a regional blanket sandstone and the primary target for carbon storage in the Illinois Basin (Figure 2), (Leetaru and McBride, 2009). The Mt. Simon sandstone is approximately 1485-ft thick at the project site and consists of stacked sand-rich braided fluvial deposits in the lower unit that transition to shelf and tidal bar sandstones in the middle and upper units (Finley et al., 2013). The lower



Figure 1. *Layout of the IBDP site. Project wells CCS1, VW1, and GM1 are indicated. The route of the CO_2 pipeline is in blue. The ADM plant is immediately south of the site. Image courtesy of the MGSC.*

System	Group	Formation
Ordovician	Maquoketa	Brainard
		Ft. Atkinson
		Scales
	Galena	Kimmswick
		Decorah
	Platteville	
	Ancell	Joachim
		St. Peter
	Prairie du Chien	Shakoppee
		New Richmond
		Oneota
		Gunter
Cambrian	Knox	Eminence
		Potosi
		Franconia
		Ironton-Galesville
		Eau Claire
		Mt. Simon
Precambrian		

Figure 2. *Stratigraphic column of Ordovician through Precambrian rocks in Illinois (Leetaru and McBride, 2009).*

Mt. Simon sandstone has average interval porosities of 20% and permeability of 185 mD and is the storage target for IBDP; however, there are low permeability units within the lower Mt. Simon Sandstone with permeabilities <1 mD that act as baffles to upward CO_2 migration (Finley et al., 2013).

The primary seal for the Mt. Simon sandstone is the regionally extensive Eau Claire shale, which is approximately 500-ft thick at the site and primarily composed of black marine shale, silty mudstones, low-permeability siltstones, and carbonates (Finley et al., 2013). The Mt. Simon sandstone is underlain by a thin Pre-Mt. Simon sandstone unit as well as granitic or rhyolitic Precambrian basement (Leetaru and McBride, 2009).

Project background

The initial site characterization work for the project took place between 2007, when the original 2D surface seismic lines were acquired, through 2010 when the final well was drilled. Early reservoir simulations suggested that, over the three-year injection period, the injected CO_2 was not predicted to move beyond the imaging aperture of a 3D VSP. Onshore 3D VSP surveys typically have better repeatability, and processed 3D VSP surveys are less affected by near-surface changes and surface noise compared to land surface seismic data. Additionally, for IBDP using time-lapse (4D) 3D VSP surveys are more economical to acquire and process than surface seismic data, so repeat VSPs were selected as the primary seismic tool to monitor the subsurface CO_2 movement.

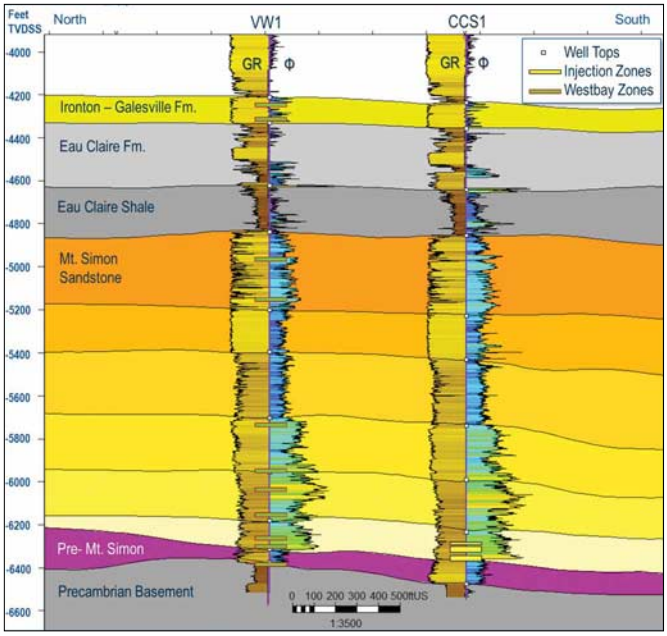


Figure 3. *Geologic cross section between CCS1 and VW1. Yellow ports in CCS1 indicate the perforated zone. Green markings in VW1 indicate the Westbay pressure, temperature, and fluid sampling ports.*

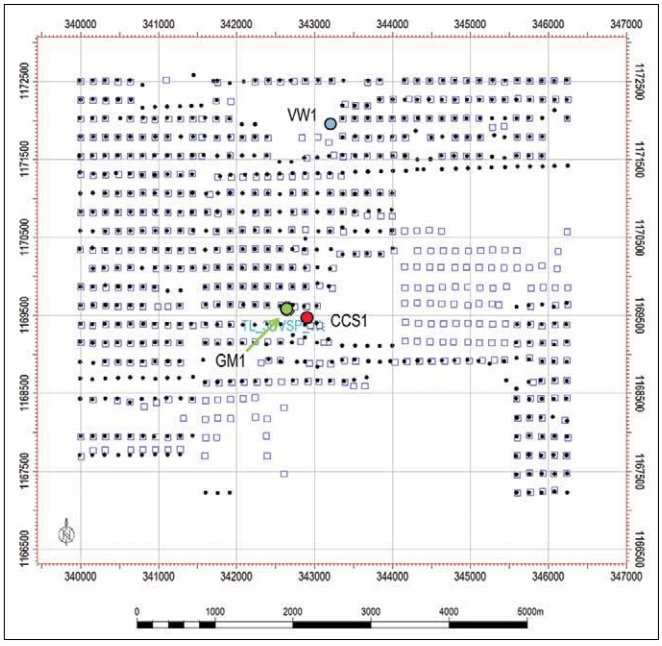


Figure 4. *Changes in shot-point geometry from Baseline 2 (B2) to Monitor 2 (M2). Blue squares represent B2 shot points. Black circles represent the M2 shot points.*

In 2008, the IBDP team installed a permanent downhole geophone array for two reasons: economics and data repeatability. It was determined to be more cost-effective to drill, complete, and install a shallow permanent downhole geophone array than to repeatedly mobilize a retrievable tool given that the project planned to acquire up to three time-lapse VSP surveys over the course of injection and one more survey post-injection. From a technical perspective, the repeatability of source and receiver geometries is a critical factor affecting

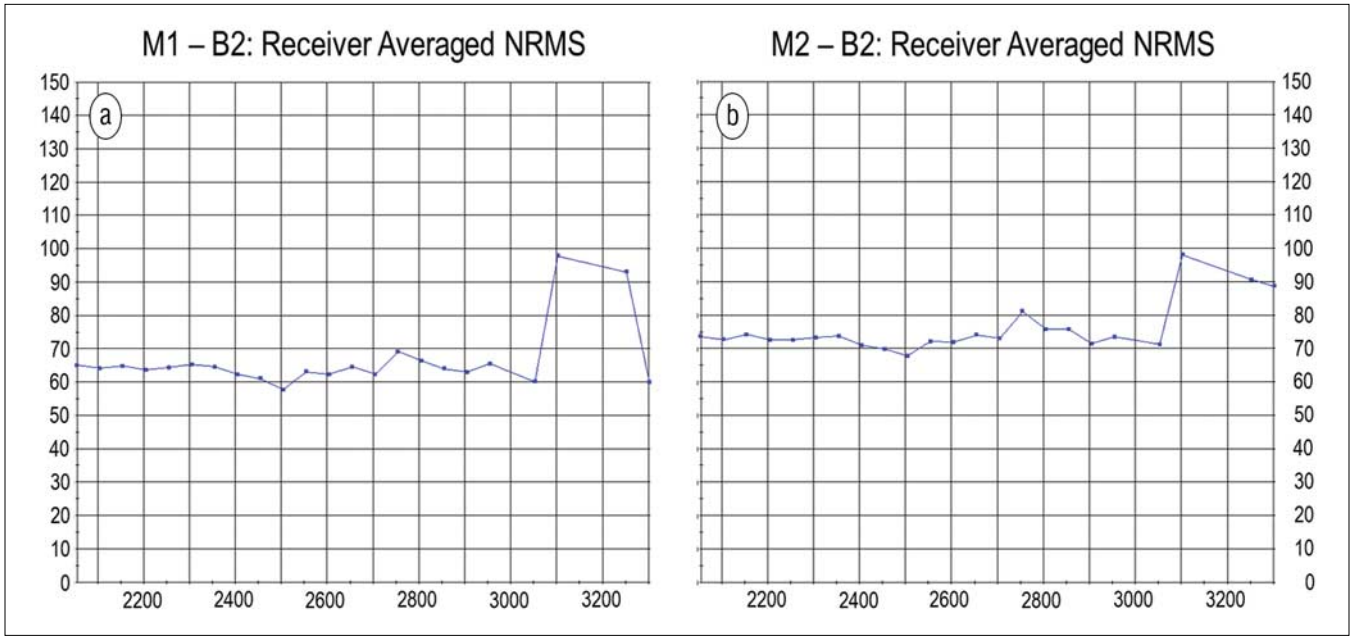


Figure 5. *(a) M1–B2 receiver averaged nrms. (b) M2–B2 receiver averaged nrms.*

the success of time-lapse seismic surveys. Landrø (1999) examined the issue of trace repeatability of offshore 3D VSP data and found that positioning errors of 15–30 ft significantly affected the repeatability of the traces. The installation of a permanent downhole geophone array removes uncertainty related to receiver positioning.

The CCS1 and GM1 wells were drilled in 2009 (Figure 1). The perforations in CCS1 are located in the lower Mt. Simon sandstone (Figure 3). GM1 is located 196 ft west of CCS1 and is instrumented with a permanent three-component 31-level geophone array that was cemented from 137–3445 ft. The VW1 well was drilled in 2010 approximately 916 ft from CCS1 and is instrumented with a Westbay multilevel groundwater characterization and monitoring system (Locke et al., 2013). The Westbay system allows real-time pressure and temperature measurements as well as discrete fluid samples to be obtained from the overlying Ironton–Galesville Formation, Mt. Simon sandstone, and the Pre-Mt. Simon unit. The pressure data from CCS1 and VW1 are used to calibrate the reservoir simulations over time.

Survey	Date	Ground Conditions	Vibrator Sweep (Hz)	Repeated Shots Relative to B2
Baseline 1	February 2010	Wet	2–100	
Baseline 2	April 2011	Dry	8–120	
Monitor 1	February 2012	Frozen/Dry	8–120	467
Monitor 2	April 2013	Damp	8–120	385

Table 1. *Comparison of the acquisition conditions and parameters for all 3D VSP surveys to-date.*

Time-lapse 3D VSP Processing Flow	
Raw Data	
60 Hz Notch Filter	
Delete Top 2 and Noisy Receiver Levels	
Linear Radon Wavefield Separation	
Downgoing Wavefield	Upgoing Wavefield
VSP Shot-by-Shot Reference Deconvolution	
Deconvolved Downgoing Wavefield	Deconvolved Upgoing Wavefield
Migration	

Table 2. *Time-lapse 3D VSP processing flow.*

Survey acquisition

The time-lapse 3D VSP surveys are being acquired with the permanent geophone array located in the GM1 to maximize receiver repeatability between surveys. The use of permanent geophone arrays for the acquisition of time-lapse 3D VSPs has not seen wide uptake (Wu et al., 2011), in part, likely because of operational logistics and concerns around well integrity. However, permanent downhole geophone arrays have the potential to provide high-resolution time-lapse images of specific targets at a lower cost than gathering surface seismic data or using retrievable VSP arrays while maintaining higher receiver repeatability over time.

Surrounded by extensive public and private cultural activity, the IBDP site is a highly challenging location for seismic

acquisition and imaging. Surface features such as housing, a community college, major industrial sites, railroads, power distribution facilities, and municipal public works infrastructure presented major challenges in the form of noise contamination, electrical interference, and obstructed access. These factors impact the quality of the time-lapse 3D VSP data that is acquired in regards to acquisition footprint and nonrepeated time-lapse noise. Over time, the footprint of the 3D VSP surveys has been further reduced through lack of access to adjacent properties (Figure 4).

Initially, the project plan was to acquire the baseline 3D VSP survey simultaneously with the 3D surface seismic data without adding additional shot points (Couëslan et al., 2008). However, survey design and modeling showed that the shot and shot-line spacing of the 3D surface seismic survey was too sparse to provide adequate 3D VSP images at the site. Since 2010, two baseline 3D VSPs and two monitor 3D VSP surveys have been acquired at IBDP (Table 1). Baseline 1 (B1) was acquired in February 2010 when the ground conditions were deteriorating as the frost started to melt, and there were also significant amounts of 60-Hz noise from electrical sources in the data. In the spring of 2011, a new baseline (B2) 3D VSP survey was acquired after work had been completed on the surface-based geophone array electronics that decreased the amount of 60-Hz noise recorded by the array. In addition, ground conditions for the B2 survey were dry, which further improved data quality.

In 2012, the first monitor 3D VSP survey (M1) was acquired after ~70,000 tons of CO_2 had been injected, which is a relatively small volume of CO_2 to detect seismically. However, the first monitor survey was timed to coincide with the first round of time-lapse RST logging in CCS1 and VW1 as well as the deep fluid sampling in VW1. The Monitor 2 3D VSP survey (M2) was acquired in April 2013 when 433,000 tons of CO_2 had been injected. Ground conditions were not as optimal as B2 and M1 because of a late snowfall and damp weather. In addition, because of inaccessibility, 77 shot points were lost on the east side of the survey.

Processing and imaging

Overall, B2 is a higher quality data set than B1 based on frequency bandwidth and signal strength, so all of the time-lapse processing has used B2 as the reference baseline data set. The primary repeatability metric used to quality control the processing was the normalized root mean square (nrms) repeatability metric (Kragh and Christie, 2001). A tolerance of 50 ft has been used for all of the surveys to interpolate and colocate shot points; approximately 90% of the shot points are collocated within 20 ft. Overall the lower nrms of the M1–B2 surveys suggests that it has better repeatability than the M2–B2 surveys. The averaged receiver nrms plots also show that the bottom four receivers had poor repeatability, so they were rejected from the time-lapse processing (Figure 5).

All of the monitor survey data was cross-equalized to B2, and the monitor survey trace amplitudes were further scaled to match the B2 survey. This was followed by standard VSP processing of wavefield separation and VSP deconvolution (Table 2). The same 1D anisotropic model that was built using ZVSP and 3D VSP traveltimes from the baseline survey was used for imaging both the data sets.

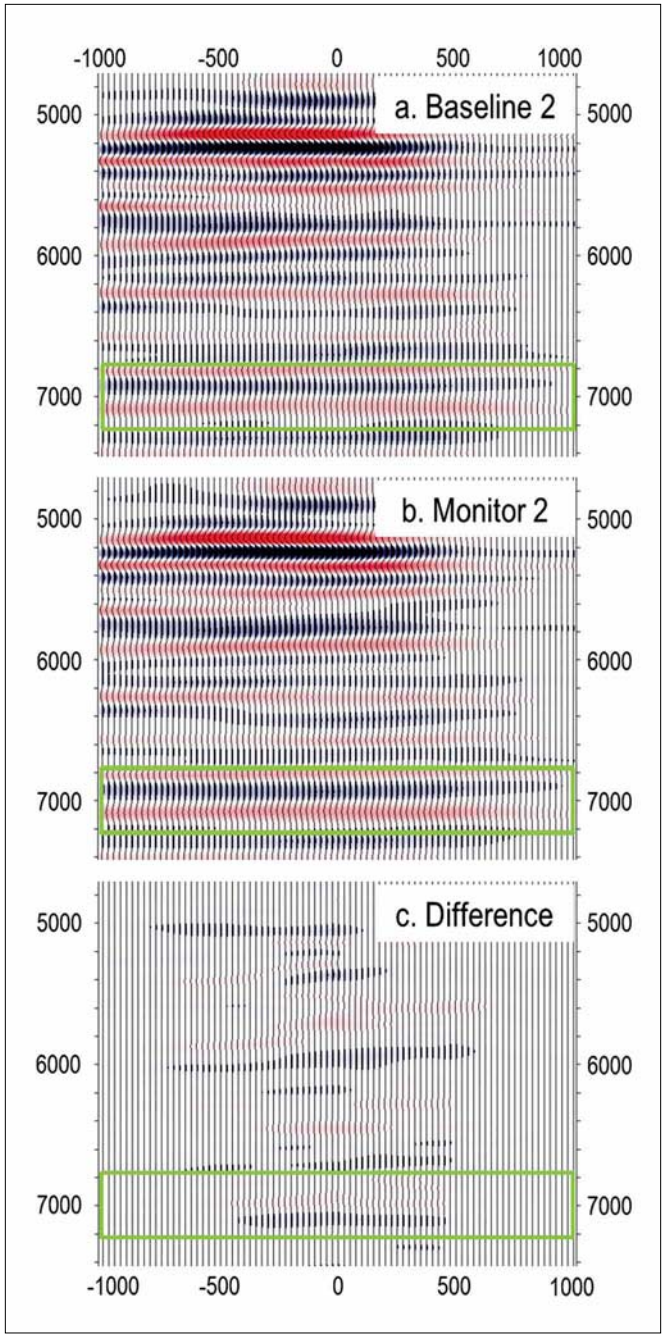


Figure 6. *(a) Migrated image from Baseline 2. (b) Migrated image from Monitor 2. (c) Difference display. The green box highlights the injection interval.*

Time-lapse 3D VSP results and correlation to other data sets

The migrated images and difference display from the B2 and M2 surveys can be seen in Figure 6. The largest potential time-lapse differences are seen at the base of and below the injection interval. Spinner logs acquired during full injection indicate that more than 80% of the CO_2 is being

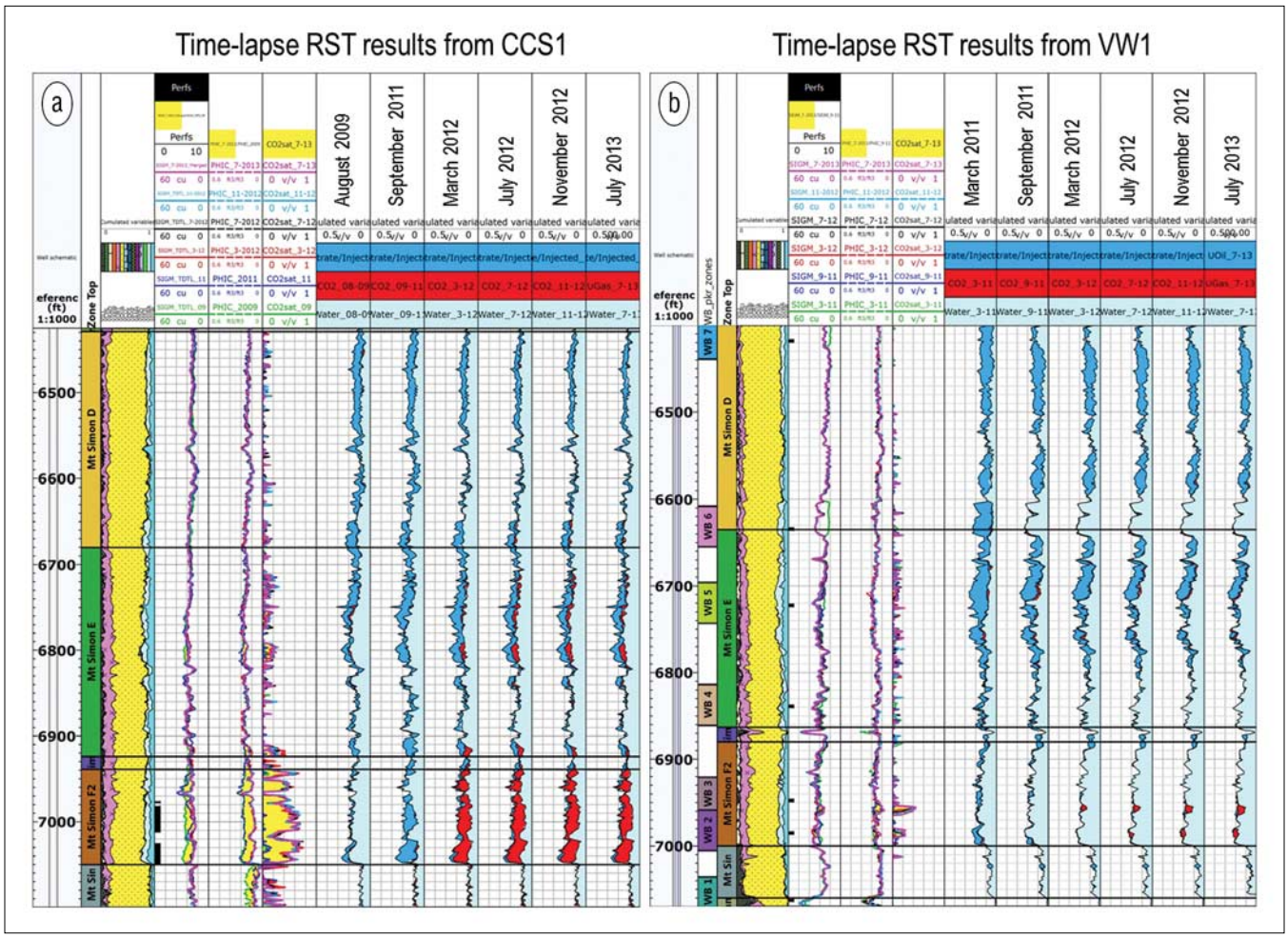


Figure 7. *(a) CCS1 time-lapse RST results. (b) VW1 time-lapse RST results. Two baseline surveys were acquired at each well. The red shading on the RST tracks denotes the CO_2. Black bars represent perforations (CCS1) and sampling points (VW1).*

injected into the bottom section of the perforated zone, and the time-lapse RST logs indicate that the bulk of the CO_2 in CCS1 is confined between 6900 and 7050 ft (Figure 7a). Therefore, it is expected that the largest time-lapse changes will be seen through this zone. While differences can clearly be seen above the injection interval (Figure 6c), the nrms metrics calculated above the injection zone indicate that these differences are not significant as the changes that have been identified through the injection interval (Figure 8).

The nrms values were calculated for two different depth ranges. The first zone was from 5000–6500 ft and encompasses all of the Eau Claire Shale and the top of the Mt. Simon sandstone; this zone should be unaffected by the CO_2 injection (Figure 8). The second interval encompasses the injection interval and ranges from 6500–7200 ft (Figure 8). Through the Eau Claire Shale and upper Mt. Simon sandstone, most of the nrms values range between 11–18% for the M1–B2 and M2–B2 surveys (Figures 8a and 8c). This demonstrates that the M1 and M2 surveys are highly repeatable with respect to B2. For the deeper interval, the time-lapse effects of CO_2 injection become apparent (Figures 8b and 8d). In both cases, the data immediately surrounding GM1 are less repeatable than the data at the western extent of the survey. In the 14 months between the M1 and M2 surveys, there has clearly been an increase in the magnitude and extent of the nrms values above 20%. This is indicative of the CO_2 plume development over time.

The vertical extent of the CO_2 plume can be further constrained through the time-lapse RST logging and pressure measurements from VW1. To date, two baseline RST surveys and four RST monitor surveys have been acquired in CCS1 and VW1. The monitor surveys were acquired after injection began in March 2012, July 2012, November 2012, and July 2013. A 6-ft interval of CO_2 was identified in VW1 in March 2012 at approximately the depth of the second perforated zone in CCS1 shortly after the M1 3D VSP survey (Finley et al., 2013). This indicated that the CO_2 plume was spreading quickly within a thin, high-permeability zone in the lower Mt. Simon sandstone. In the July 2013 RST logging runs, CO_2 was observed in CCS1 between 6750 and 7048 ft; however, most of the CO_2 appears to be confined below 6900 ft (Figure 7). In VW1, CO_2 has been identified between 6943 and 6990 ft and is still confined to a couple of relatively thin intervals (Figure 7).

Pressure measurements from VW1 further support the fact that the CO_2 is confined below 6837 ft. Pressures measured at Westbay sampling ports 2 and 3, at 6982 and 6945 ft, respectively, respond quickly to injection operations. However, the pressures at Westbay sampling port 4 (6837 ft) have shown only a small and gradual increase over time indicating that the CO_2 plume has not migrated to this depth. A low-permeability zone in the lower Mt. Simon sandstone is known to exist between sampling ports 3 and 4 based on the well-log data, and it appears to be blocking pressure and fluid transmission at this time.

Discussion

The nrms repeatability metrics through the injection interval show stable anomalies, which are indicative of CO_2 plume development between the B2, M1, and M2 3D VSP surveys. However, more work is required to support the interpretation of these observations. The time-lapse seismic response in rock/fluid systems is a complex combination of rock and fluid properties that change over time and space. This can cause ambiguity in the direct interpretation of the seismic data on a stand-alone basis. This complexity increases in CO_2-brine systems where additional processes such as phase transitions, dehydration, and changes to the rock matrix may occur through mineral dissolution and precipitation. Some of these ambiguities may be significantly reduced through the addition of rock physics and fluid substitution modeling to the interpretation workflow. Rock physics modeling has been performed using the log data from the IBDP wells; however, the fluid substitution modeling will be revisited using the new pressure and saturation predictions derived from the latest calibrated reservoir simulation model. Pending the results of the new fluid substitution modeling, the interpretation of the time-lapse seismic response may be used to update the spatial plume predictions from the reservoir model in regions lacking direct MVA measurements.

Based on the results of the M2 survey, IBDP will likely proceed with a third monitor survey in 2014 near the end of injection because the project may be required to demonstrate that the bulk of the CO_2 plume can be identified through MVA data. The results of the IBDP time-lapse 3D VSPs will also be used to help inform the MVA plan that is being developed for the adjacent Illinois Industrial CCS Project (ICCS). While the Illinois ICCS Project plans to inject higher volumes of CO_2, time-lapse 3D VSPs will likely still play a large role in monitoring CO_2 plume development over the life of that project as well. TLE

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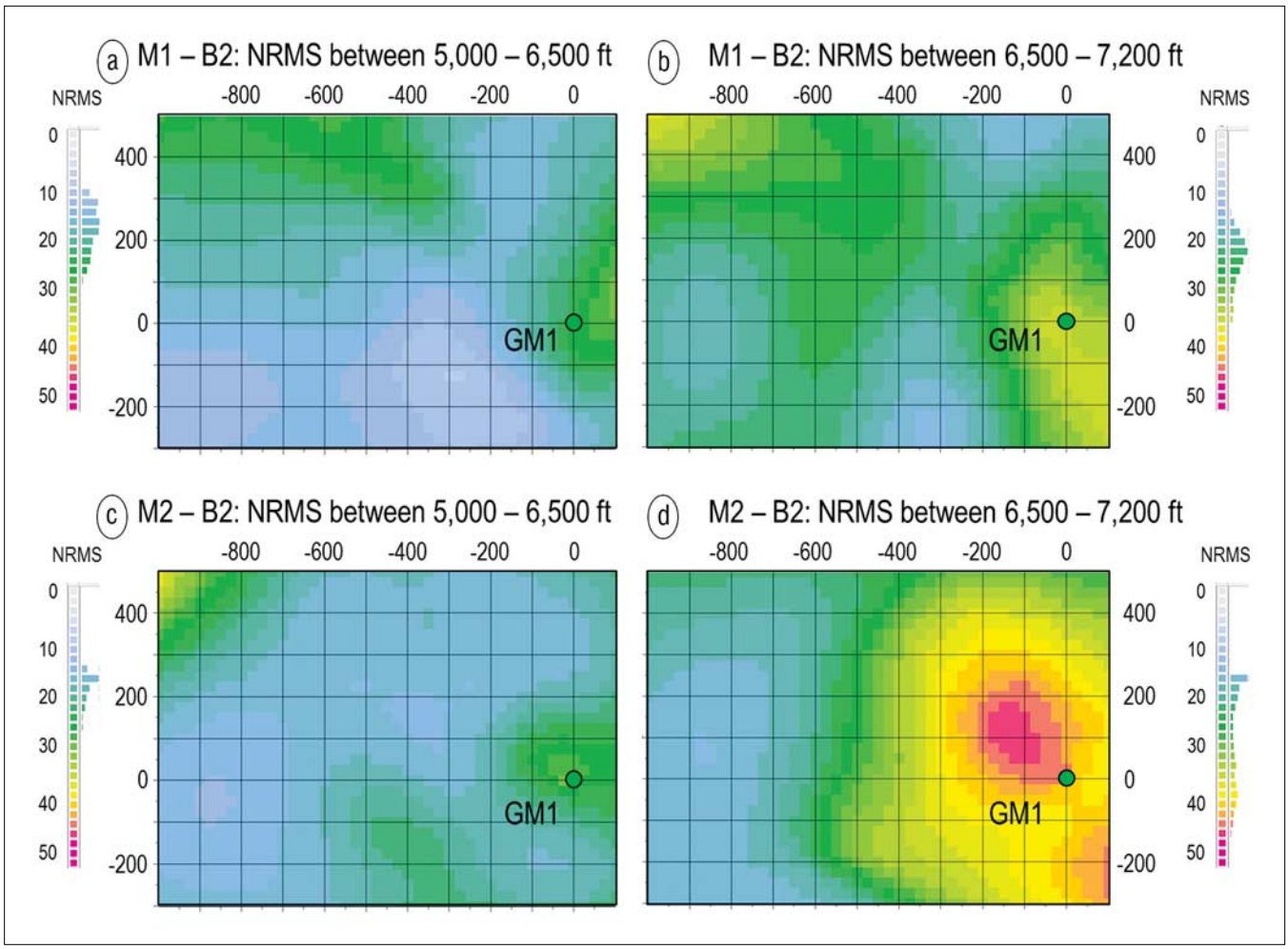


Figure 8. *(a) M1–B2 nrms values through the Eau Claire shale and upper Mt. Simon sandstone. (b) M1–B2 nrms values over a zone encompassing the injection interval. (c) M2–B2 nrms values through the Eau Claire shale and upper Mt. Simon sandstone. (d) M2–B2 nrms values over a zone encompassing the injection interval.*

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